

EMAT 102**B.Tech. Ist SEMESTER EXAMINATION, 2024-25****BACHELOR OF TECHNOLOGY****(CSE & ME)****Engineering Mathematics II**

Time : Three Hours]

[Maximum Marks : 75

Note: There are **three** sections (A, B and C) and Candidate has to attempt questions from all sections. Marks are indicated against each section.

Section-A

1. Attempt all questions. Each question carries equal marks :

 $3 \times 5 = 15$

- (a) Define order of differential equations and find the order and degree of –

$$\frac{d^2y}{dx^2} = \left(1 + \frac{dy}{dx}\right)^{4/5}$$

- (b) Solve : $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$.

- (c) State first and second shifting property of Laplace transform.

- (d) Find inverse Laplace transform of $\frac{1}{s(s+1)}$.

- (e) Solve : $\sqrt{p} + \sqrt{q} = 1$.

Section-B

Note : Attempt all questions. Each questions carries equal marks :

$5 \times 4 = 20$

2. (a) Solve : $(D^2 - 2D + 3)y = x^3 + \sin x$.

Or

(b) Solve : $(D^2 + 2D + 10)y + 37 \sin 3x = 0$.

3. (a) Solve : $\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)}$.

Or

(b) Evaluate : $\int_0^\infty \frac{e^{-2t} \sin 2t}{t} dt$.

4. (a) If $L[F(t)] = f(s)$, then prove that :

$$L[F'(t)] = s f(s) - F(0)$$

Or

(b) Find inverse Laplace transform of $\frac{1}{s^2 + s + 1}$

5. (a) Find the Fourier series of the function :

$$f(x) = |x|, -\pi < x < \pi.$$

Or

(b) Find the Fourier series of :

$$f(x) = x^2, -2 \leq x \leq 2.$$

Section-C

Note : Answer any two questions of the following : $2 \times 20 = 40$

6. Using variation of parameter, to solve

$$\frac{d^2y}{dx^2} + a^2y = \sec ax .$$

7. Using Laplace transform to solve the differential equation :

$$y''(t) + 25y(t) = 10\cos 5t, y(0) = 2, y'(0) = 0$$

8. Solve the equation by using separation of variables.

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u, \text{ given } u(x, 0) = 6e^{-3x}$$

9. Find the Fourier series expansion of –

$$f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ x, & 0 < x < \pi \end{cases}$$

Hence find the sum of series :

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

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